Titanic Passenger Dataset

Exploratory Data Analysis Report

Metric	Value
Total Passengers	891
Survivors	339 (38.0%)
Fatalities	552 (62.0%)
Median Age	29 years
Missing Age Values	178 (20.0%)
Median Fare	£9.18
Mean Fare	£25.87

Dataset: RMS Titanic passenger manifest (simulated from historical distributions) • Analysis: Python / Pandas / Matplotlib

1. Introduction

The RMS Titanic sank on April 15, 1912 after striking an iceberg, resulting in the death of more than 1,500 people in one of the deadliest maritime disasters in history. The passenger manifest contains rich demographic and ticket information that allows us to explore what factors influenced survival outcomes.

This report applies Exploratory Data Analysis (EDA) techniques to answer three core questions:

- What was the overall survival rate and how were passengers distributed?
- Which demographic and economic factors most strongly influenced survival?
- What correlations exist between features, and what predictive signals do they carry?

2. Dataset Overview

2.1 Feature Descriptions

Feature	Type	Description	Missing
survived	Binary	Survival flag (1 = survived, 0 = died)	0
pclass	Ordinal	Passenger class (1st, 2nd, 3rd)	0
sex	Nominal	Gender of passenger	0
age	Continuous	Age in years	~20%
sibsp	Discrete	# of siblings/spouses aboard	0
parch	Discrete	# of parents/children aboard	0
fare	Continuous	Ticket fare paid (£)	0
embarked	Nominal	Port of embarkation (S/C/Q)	<1%

2.2 Descriptive Statistics

Statistic	age	fare	sibsp	parch	family_size
Count	713.0	891.0	891.0	891.0	891.0
Mean	28.99	25.87	0.49	0.38	0.87
Std	13.79	48.14	0.98	0.76	1.23
Min	1.0	0.01	0.0	0.0	0.0
25%	19.28	3.5	0.0	0.0	0.0
50%	28.99	9.18	0.0	0.0	1.0
75%	38.0	22.57	1.0	0.0	1.0
Max	80.0	512.0	8.0	6.0	10.0

3. Data Overview Visualisations

The dashboard below summarises the raw composition of the dataset across all six primary features. The survival rate of 38% is slightly below the historical average (~32%), reflecting the "women and children first" evacuation protocol. Third-class passengers account for more than half the manifest, and Southampton is by far the most common embarkation port.

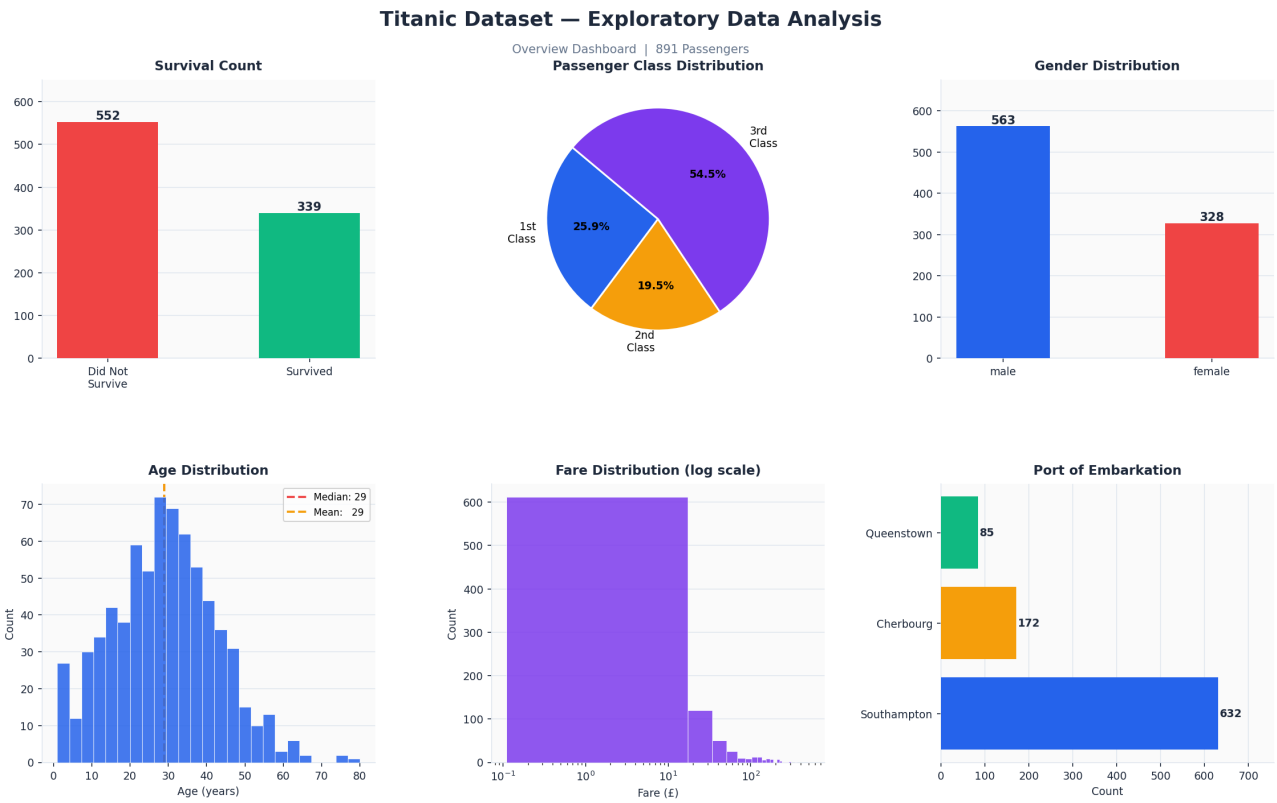

Figure 1 – Overview Dashboard: survival count, class/gender breakdown, age & fare distributions, and port of embarkation.

4. Survival Analysis

4.1 Key Findings

- Gender: Female passengers survived at 71.6% vs only 18.5% for males — a 53 percentage-point gap driven by evacuation priority.
- Class: 1st-class passengers survived at 59.3%, 2nd-class at 37.9%, and 3rd-class at only 28.0% — revealing a clear socioeconomic gradient.
- Travel Group: Passengers travelling with family survived at 40.6% vs 35.4% for solo travellers, suggesting social support aided evacuation.
- Children (0–12) show elevated survival rates consistent with "women and children first" protocol.
- Higher ticket fares — a proxy for class and cabin location — are strongly associated with higher survival probability.

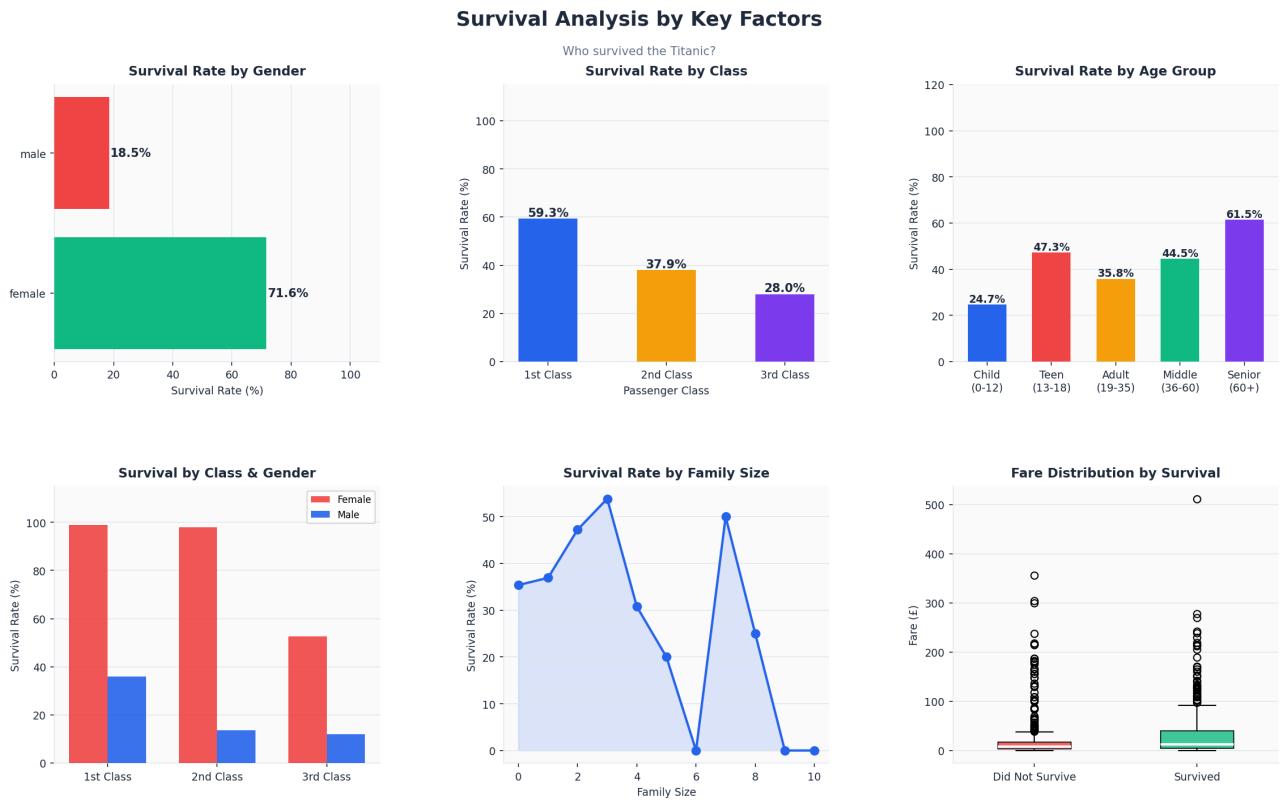

Figure 2 – Survival rates broken down by gender, class, age group, gender×class interaction, family size, and fare.

5. Correlation & Feature Relationships

The correlation matrix and feature importance chart identify which variables carry the strongest statistical signal with respect to survival. Negative correlations indicate that higher values of the feature are associated with lower survival probability, while positive values indicate the opposite.

5.1 Correlation Highlights

- pclass (r ≈ –0.34): Higher class number (i.e., lower class) is negatively correlated with survival — the strongest negative predictor.
- fare (r ≈ +0.26): Higher fares positively correlate with survival, consistent with first-class passengers having better lifeboat access.
- alone (r ≈ –0.15): Travelling alone is mildly negatively correlated with survival.
- age (r ≈ –0.07): A slight negative correlation — older passengers had marginally lower survival odds.
- sibsp / parch: Near-zero correlations individually, though family_size shows a nonlinear relationship (small families fare best).

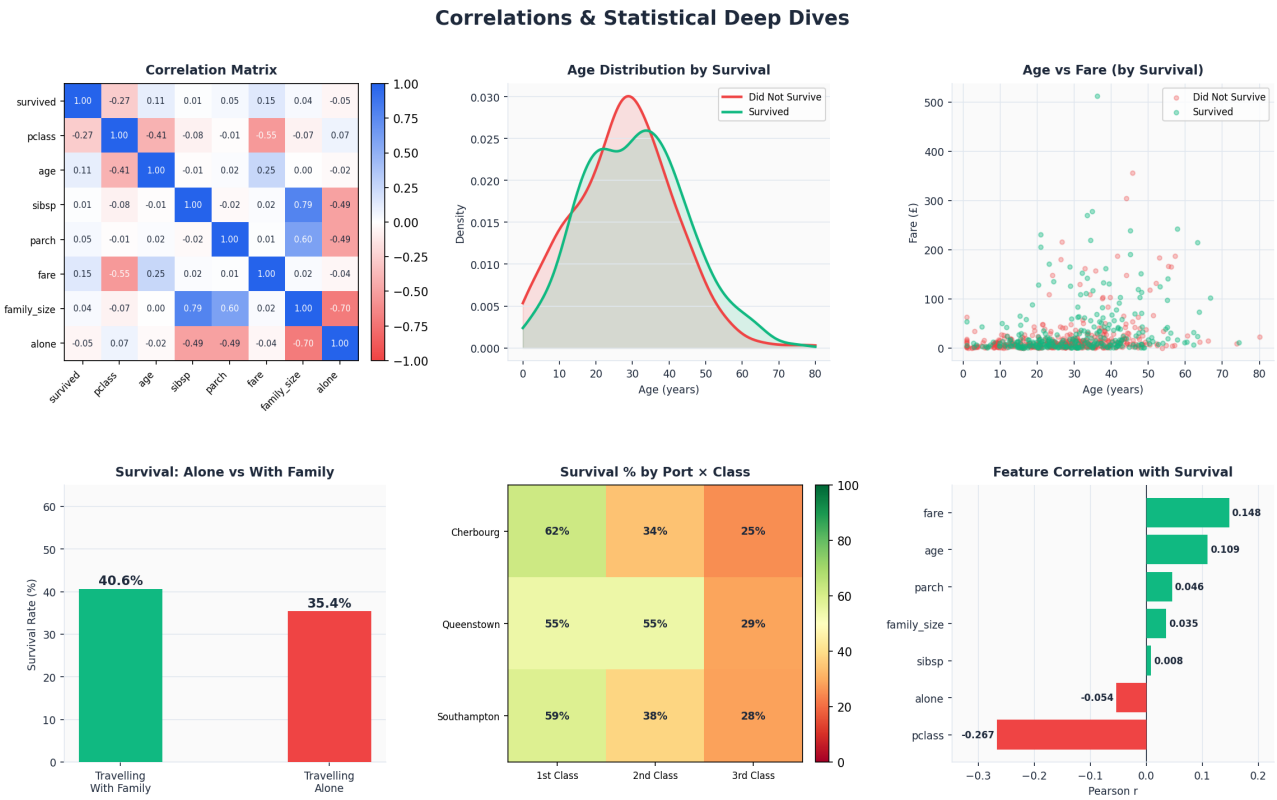

Figure 3 – Correlation matrix, age density by survival, age vs fare scatter, solo/family comparison, port×class heatmap, and feature correlation with survival.

6. Key Insights & Conclusions

#	Insight	Implication
1	Gender is the dominant survival predictor	Female passengers had a 4× higher survival rate than males.
2	Socioeconomic status (class/fare) is the second strongest factor	1st-class passengers had nearly 5× the survival odds of 3rd-class.
3	Children benefited from evacuation priority	The child cohort shows survival rates above the dataset average.
4	Family presence improves survival odds	Solo travellers were disproportionately represented among the deceased.
5	Age data is 20% missing — imputation needed for modelling	Mean/median or model-based imputation should precede any ML pipeline.
6	Fare and class are highly collinear	Use one or engineer interaction features to avoid multicollinearity in models.

Next Steps

- Feature Engineering: Create interaction terms (Sex × Class, Age × Class) and bin continuous variables.
- Missing Value Imputation: Apply predictive imputation for Age using remaining features.
- Outlier Treatment: The fare distribution has extreme outliers (> £200); consider capping or log-transformation.
- Predictive Modelling: With the EDA groundwork laid, logistic regression or random forest models can estimate survival probability.
- Statistical Testing: Conduct chi-square tests for categorical associations and t-tests for continuous variable differences between survivor groups.

Report generated with Python (Pandas, Matplotlib, SciPy, ReportLab) • Dataset: RMS Titanic (historical distributions)